Home / Grade 8 / Geometry / Quarter 4

The inscribed angle

Move the vertex anywhere along the arc and the angle never changes — the most useful theorem in the chapter.

LESSON 56 | 1 × 40 MIN | GEOMETRY 8, TEMA 57 | STAGE 9 · 7.1

LESSON CLOCK

5' | 14' | 12' | 5' | 4'

Guess before measuring	5 min
The theorem	14 min
The three corollaries	12 min
Cyclic quadrilaterals	5 min
Homework	4 min

LEARNING OBJECTIVES

- State and use the inscribed angle theorem.
- Use the corollary that an angle in a semicircle is a right angle.
- Use the corollary that angles on the same arc are equal.
- Find angles in a cyclic quadrilateral.

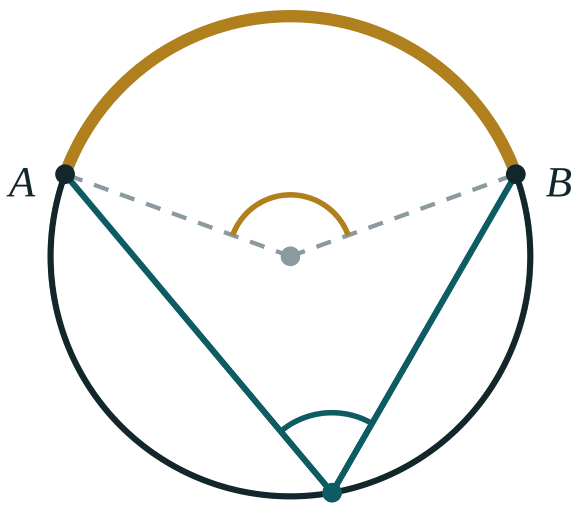
TEXTBOOK

National	Тема 57, pp. 135–137
Cambridge	Learner’s Book pp. 142–148
Workbook	Workbook 7.1

01 Explanation

The theorem

An angle with its vertex **on** the circle, whose arms are two chords, is an **inscribed angle**.



∠ACB = ½C∠AOB

An inscribed angle is exactly half the central angle standing on the same arc.

INSCRIBED ANGLE THEOREM

$$\angle ACB = \frac{1}{2}\ \angle AOB$$

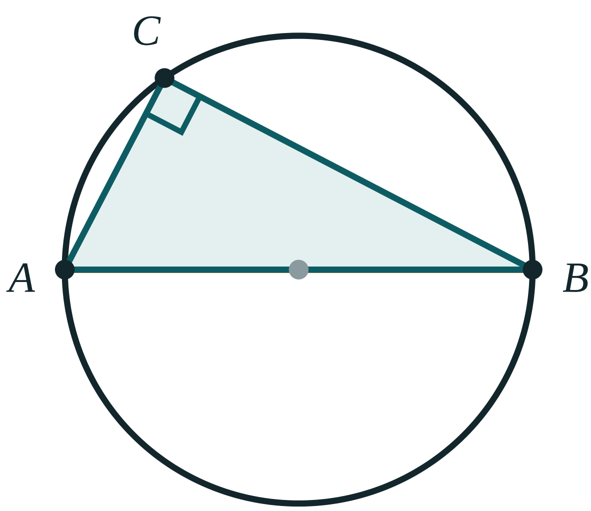
An inscribed angle is **half** the central angle on the same arc — equivalently, half the degree measure of that arc.

The proof, in the case where C, O and one chord line up: triangle OCB is isosceles ($OC = OB = r$), so its base angles are equal, say β each. The exterior angle at O is then 2β — and that exterior angle is the central angle. The general case follows by splitting the angle into two such pieces.

Three corollaries you will use constantly

1 · ANGLES ON THE SAME ARC

All inscribed angles standing on the **same arc** are equal — they are all half of the one central angle. Move the vertex anywhere along the far arc; the angle does not move.



an angle on a diameter is always 90°

An angle inscribed in a semicircle is always a right angle, because the central angle is the straight angle 180°.

2 · ANGLE IN A SEMICIRCLE

An angle subtended by a **diameter** is $90°$ — the central angle is the straight angle $180°$, and half of that is $90°$.

3 · CYCLIC QUADRILATERAL

In a quadrilateral whose four vertices lie on one circle, opposite angles add to $180°$:

$$\angle A + \angle C = \angle B + \angle D = 180°$$

because the two arcs they stand on make up the whole $360°$.

SAME ARC, NOT JUST SAME CHORD

Two angles standing on the same chord but on **opposite sides** of it are not equal — they are supplementary. That is corollary 3 in disguise.

02 Worked examples

EXAMPLE 1 **A central angle is $80°$. Find the inscribed angle on the same arc.**

1. $\angle inscribed = \frac{1}{2} \cdot 80°$

2. $= 40°$

   And every inscribed angle on that arc is $40°$.

ANSWER

$40°$

EXAMPLE 2 **AB is a diameter and C is on the circle, with $\angle CAB = 35°$. Find the other two angles of triangle ABC.**

1. $\angle ACB = 90°$

   Angle in a semicircle.

2. $\angle ABC = 180° - 90° - 35°$

   Angle sum of a triangle.

3. $= 55°$

ANSWER

$\angle ACB = 90°$, $\angle ABC = 55°$

EXAMPLE 3 **In a cyclic quadrilateral $ABCD$, $\angle A = 78°$ and $\angle B = 95°$. Find $\angle C$ and $\angle D$.**

1. $\angle C = 180° - 78° = 102°$

   Opposite to $\angle A$.

2. $\angle D = 180° - 95° = 85°$

   Opposite to $\angle B$.

3. $78 + 95 + 102 + 85 = 360$

   Check ✓

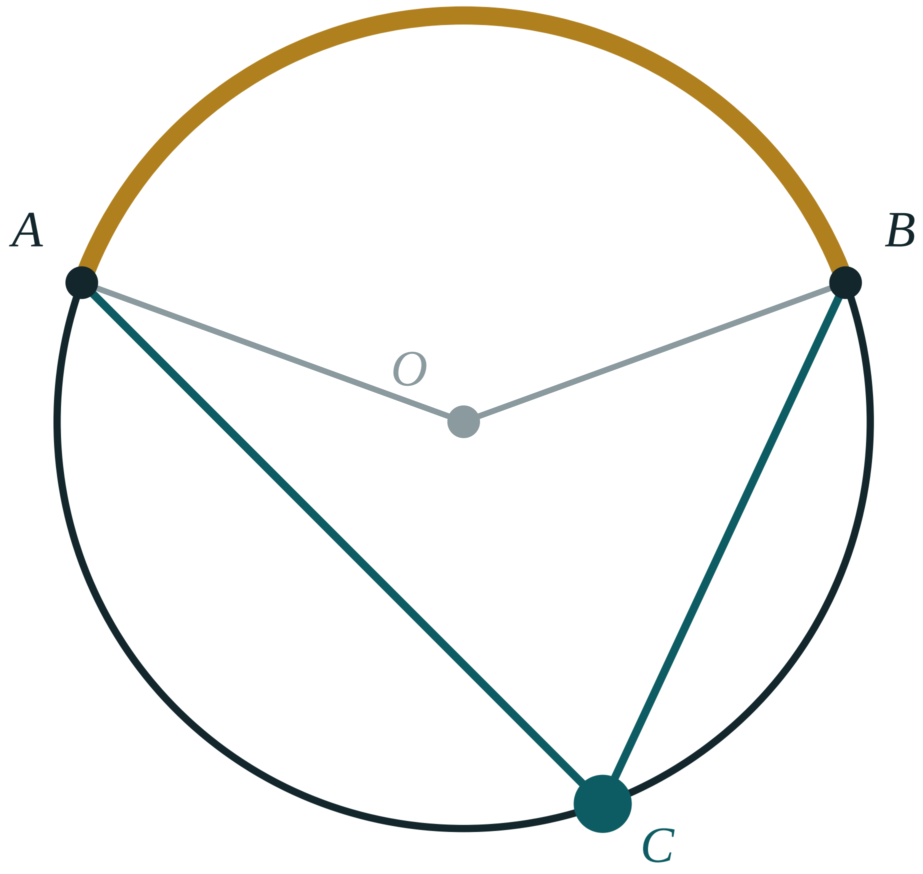
ANSWER

$\angle C = 102°$, $\angle D = 85°$

03 Interactive model

Drag C round the circle: the inscribed angle stays exactly half the central angle.

Inscribed and central angles



central ∠AOB **140°** | inscribed ∠ACB **70°** | ratio **2** | arc AB **140°**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Inscribed angle	Ichki chizilgan burchak	Вписанный угол
Subtended by an arc	Yoyga tiralgan	Опирающийся на дугу
Angle in a semicircle	Yarim aylanaga tiralgan burchak	Угол, опирающийся на диаметр
Same arc	Bir xil yoy	Одна и та же дуга
Cyclic quadrilateral	Aylanaga ichki chizilgan to‘rtburchak	Вписанный четырёхугольник
Opposite angles	Qarama-qarshi burchaklar	Противоположные углы
Corollary	Natija	Следствие
Diameter	Diametr	Диаметр

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

A central angle of $120°$ gives an inscribed angle of:

- $60°$
- $120°$
- $240°$
- $30°$

An angle inscribed in a semicircle is:

- $45°$
- $60°$
- $90°$
- $180°$

In a cyclic quadrilateral, $\angle A = 110°$. Then $\angle C$ is:

- $110°$
- $70°$
- $90°$
- $250°$

Two inscribed angles on the same arc are:

- equal
- supplementary
- complementary
- unrelated

An inscribed angle of $25°$ stands on an arc of:

- $12.5°$
- $25°$
- $50°$
- $75°$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. *A central angle is 60°. Find the inscribed angle on the same arc.*

   ANSWER *30°*

2. *An inscribed angle is 40°. Find the central angle.*

   ANSWER *80°*

3. *AB is a diameter and C is on the circle. Find ∠ACB.*

   ANSWER *90°*

4. *In a cyclic quadrilateral, ∠B = 85°. Find ∠D.*

   ANSWER *95°*

5. *An inscribed angle stands on a 100° arc. Find it.*

   ANSWER *50°*

6. *An inscribed angle is 30°. Find its arc.*

   ANSWER *60°*

7. *Two inscribed angles stand on the same arc and one is 47°. Find the other.*

   ANSWER *47°*

Medium · the standard question

7 PROBLEMS

1. *AB is a diameter and $\angle CAB = 28°$. Find $\angle ABC$.*

   ANSWER $62°$

2. *In a cyclic quadrilateral, $\angle A = 3\angle C$. Find both.*

   ANSWER $\angle A = 135°$, $\angle C = 45°$

3. *A central angle is $140°$. Find the inscribed angle on the major arc.*

   ANSWER $70°$

4. *An inscribed angle of $65°$ stands on arc AB. Find the central angle AOB.*

   ANSWER $130°$

5. *In a cyclic quadrilateral ABCD, $\angle A = 92°$ and $\angle D = 71°$. Find $\angle B$ and $\angle C$.*

   ANSWER $\angle C = 88°$, $\angle B = 109°$

6. *AB is a diameter, $\angle ABC = 40°$. Find the arc AC.*

   ANSWER $80°$

7. *An inscribed angle stands on a semicircle. What is its arc?*

   ANSWER $180°$

Hard · stretch and reason

7 PROBLEMS

1. *AB is a diameter of a circle of radius 10 and $\angle CAB = 30°$. Find BC and AC.*

   ANSWER $BC = 10$, $AC = 10\sqrt{3} \approx 17.3$

2. *In a cyclic quadrilateral the angles, taken in order, are in the ratio 2 : 3 : 4 : 3. Find them.*

   ANSWER $60°, 90°, 120°, 90°$ — opposites: $60 + 120 = 180$ and $90 + 90 = 180$ ✓

3. *Two chords AB and CD meet at P inside a circle. Show that triangles APC and DPB are similar.*

   ANSWER $\angle A = \angle D$ (same arc BC) and $\angle C = \angle B$ (same arc AD), so the triangles have two equal angles each.

4. *A triangle inscribed in a circle has one side as a diameter of length 26 and another side 10. Find the third side.*

   ANSWER 24 — the angle opposite the diameter is $90°$

5. *An inscribed angle of 35° and a central angle on the same arc are drawn. Find the reflex central angle.*

   ANSWER $360° - 70° = 290°$

6. *Prove that opposite angles of a cyclic quadrilateral add to 180°.*

   ANSWER $\angle A$ and $\angle C$ are inscribed on the two arcs that together make the whole circle, so their sum is $\frac{1}{2} \cdot 360° = 180°$.

7. *A chord of a circle of radius 8 subtends a central angle of 90°. Find its length and the inscribed angle on the major arc.*

   ANSWER $8\sqrt{2} \approx 11.3$, inscribed angle $45°$

07 Homework

Homework — 5 problems

Geometry 8, Тема 57, pp. 135–137. Name the arc every angle stands on.

1. *A central angle is $110°$. Find the inscribed angle on the same arc.*
2. *AB is a diameter and $\angle CAB = 42°$. Find the other two angles of triangle ABC.*
3. *In a cyclic quadrilateral, $\angle A = 105°$ and $\angle B = 68°$. Find $\angle C$ and $\angle D$.*
4. *An inscribed angle of $55°$ stands on arc AB. Find the arc and the central angle.*
5. *A triangle inscribed in a circle has a diameter of 20 as one side and another side 12. Find the third side.*